

Backend Development Proposal – Milestone 3

1. Executive Summary

This proposal outlines the comprehensive backend development services for Milestone 3, covering the implementation of **Products**, **Services**, and **Learning Management System (LMS)** modules. The system will be developed using Node.js and Express, following best practices in scalability, security, and modular design.

The implementation will strictly follow the schema definitions provided, ensuring seamless frontend integration and long-term maintainability. All development will be supported by complete, professional-grade documentation.

2. Scope of Work

A. Products Module

Endpoint	Method	Description
/products/all	GET	Retrieves a list of all products (Super Admin access only)
/admin/products	GET	Displays admin panel for product management
/admin/products/create	POST	Creates a new product entry
/admin/products/update	POST	Updates an existing product's details

Module Features:

- Admin panel with product CRUD operations
 - Schema-conformant data handling
 - Access restricted to Super Admin users
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B. Services Module

Endpoint	Method	Description
/services/{category}	GET	Returns services based on user category and location coordinates

Module Features:

- Google Places API integration for business listings
- Location-aware filtering via latitude and longitude
- Caching mechanism to minimize external API calls
- Rate limiting and user authentication

C. Learning Management System (LMS) Module

Endpoint	Method	Description
/lms/classes/all	GET	Fetches all available learning modules
/admin/lms	GET	Displays admin interface for LMS content management
/admin/lms/class/create	POST	Allows creation of LMS classes with markdown and assets
/admin/lms/class/update	POST	Enables updating of LMS classes and content assets

Module Features:

- Admin dashboard to manage learning content
- Markdown-based course creation
- File/image upload support
- Structured schema compliance
- Public access configuration for content assets

Additional Functionalities

- **Bulletin Board** **Module**
A non-real-time messaging component for administrators to post announcements or updates.
 - **Image Upload** **Handling**
Backend support for image and asset uploads linked to products and LMS classes.
 - **Security Layer**
 - Token-based authentication
 - Rate limiting
 - Secure input validation and request handling
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3. Deliverables

Deliverable	Description
Backend APIs	Complete API implementations for Products, Services, and LMS modules
Admin Interfaces	Dashboard views for authorized users to manage data
Google API Integration	Service discovery feature utilizing Google Places
Upload Infrastructure	Image and file upload endpoints and storage logic
Security Enhancements	Authentication, rate limiting, and security controls
Professional Documentation	

- Full route documentation
- Sample request and response objects
- API usage guides
- Setup and deployment instructions
- Technical commentary on key logic components

All documentation will be delivered in Markdown and PDF formats for use by developers, QA, and operations teams.

4. Timeline

Phase	Task	Estimated Duration
Phase 1	Product Module Implementation	4–5 working days
Phase 2	Services Module with Google Integration	3–4 working days
Phase 3	LMS Module with Upload Functionality	4–5 working days
QA & Documentation	Testing, review, and full documentation	3–4 working days

Total Estimated Timeline: 14 to 18 working days

This range accounts for complexity variations and incorporates time for internal review, refinements, and feedback.

5. Technology Stack

Component	Technology
Backend Framework	Node.js with Express
Authentication	JSON Web Tokens (JWT)
API Integration	Google Places API
Uploads	Multer or equivalent file handler
Documentation	Swagger, Postman, Markdown, or PDF export

6. Cost Summary

Item	Cost (USD)
Product Module Development	150
Services API and Google Integration	100
LMS Module with Upload & Markdown Support	150
Admin Dashboards (Product & LMS)	50
Testing and Professional Documentation	50

Total Cost: USD 500

7. Conclusion

This proposal outlines a complete backend implementation that aligns with the existing documentation, technical requirements, and structural guidelines already in place. Our development approach is centered on quality, clarity, and long-term maintainability.

Upon approval, we will proceed with a structured, milestone-based execution plan and provide consistent progress updates throughout the project.

Please let me know if anything is unclear or if you'd like any modifications to the proposal. I'm happy to make adjustments based on your feedback. We look forward to delivering a reliable, professional backend system tailored to your operational and product goals.